Question ID: 8584f3ce

Question

When digging for clams, their primary food, sea otters damage the roots of eelgrass plants growing on the seafloor. Near Vancouver Island in Canada, the otter population is large and well established, yet the eelgrass meadows are healthier than those found elsewhere off Canada's coast. To explain this, conservation scientist Erin Foster and colleagues compared the Vancouver Island meadows to meadows where otters are absent or were reintroduced only recently. Finding that the Vancouver Island meadows have a more diverse gene pool than the others do, Foster hypothesized that damage to eelgrass roots increases the plant's rate of sexual reproduction; this, in turn, boosts genetic diversity, which benefits the meadows' health overall.

Which finding, if true, would most directly undermine Foster's hypothesis?

Answer

A. At some sites in the study, eelgrass meadows are found near otter populations that are small and have only recently been reintroduced.

B. At several sites not included in the study, there are large, well-established sea otter populations but no eelgrass meadows.

C. At several sites not included in the study, eelgrass meadows' health correlates negatively with the length of residence and size of otter populations.

D. At some sites in the study, the health of plants unrelated to eelgrass correlates negatively with the length of residence and size of otter populations.